



PENGENALAN PLC GE PACSYSTEM RX3I

M. Yazid

30 Juli 2011

Wina Gresik



GE-Fanuc / GE-IP

- Series 90™-30
- Series 90-70
- VersaMax® I/O & Control
- VersaMax Nano/Micro
- PACSystems™RX3i
- PACSystems RX7i

Products



Series 90-30



Series 90-70

Products



Versamax Modules



Products



Versamax Nano

Versamax Micro



Products

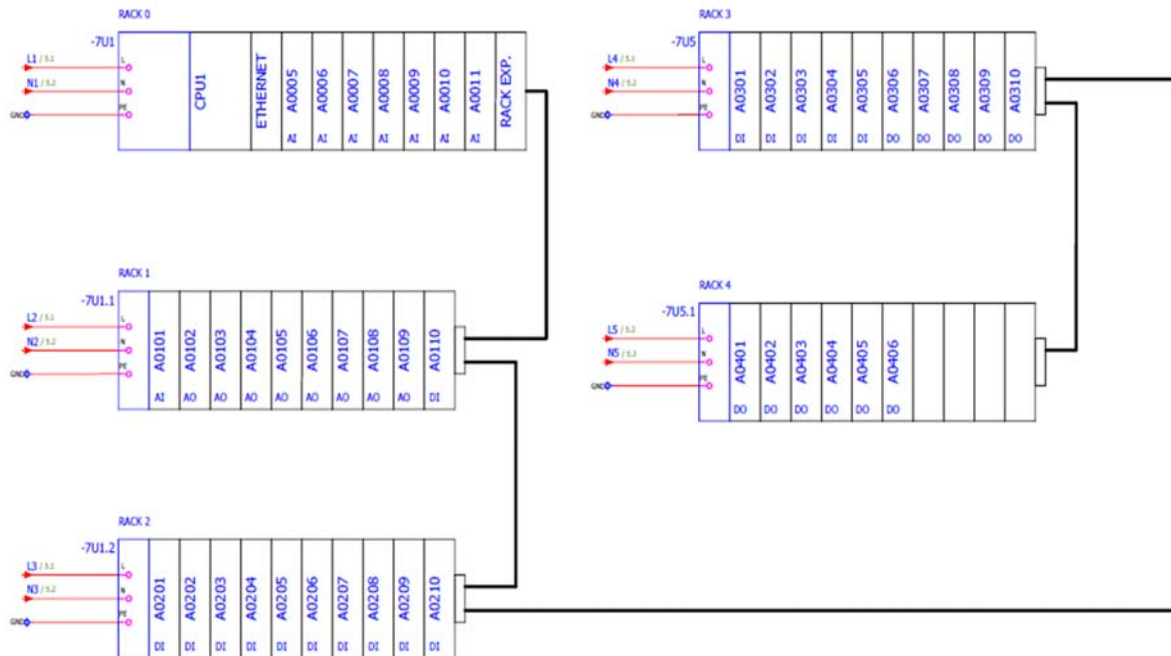


PACSystem Rx7i

PACSystem Rx3i



PLC System



PLC Parts

Beberapa bagian PLC sistem yang paling umum antara lain :

- **Backplane**
- **Power Supply**
- **Serial Bus Transmitter**
- **Discrete/Digital Input Module**
- **Discrete/Digital Output Module**
- **Analog Input Module**
- **Analog Output Module**

Selain itu ada :

- **Analog Module with Hart Communication**
- **Discrete/Digital Mixed Module**
- **Analog Mixed Module**
- **Special Purpose Module**



Backplane

Backplane digunakan sebagai tempat di mana power supply dan module-module yang lain dipasang.

Rx3i memiliki beberapa jenis backplane, yaitu :

- **16 Slot RX3i Universal Backplane, IC695CHS016**
- **12 Slot RX3i Universal Backplane, IC695CHS012**
- **5 Slot RX3i Serial Expansion Backplane, IC694CHS398**
- **10 Slot RX3i Serial Expansion Backplane, IC694CHS392**

Selain itu beberapa backplane di bawah ini bisa digabung dengan sistem Rx3i

Backplane Type	Catalog Number
16 Slot RX3i Universal Backplane	IC695CHS016
12 Slot RX3i Universal Backplane	IC695CHS012
5 Slot RX3i Serial Expansion Backplane	IC694CHS398
10 Slot RX3i Serial Expansion Backplane	IC694CHS392
5 Slot Series 90-30 Expansion Backplane	IC693CHS398
10 Slot Series 90-30 Expansion Backplane	IC693CHS392
5 Slot Series 90-30 Remote Backplane	IC693CHS399
10 Slots Series 90-30 Remote Backplane	IC693CHS393

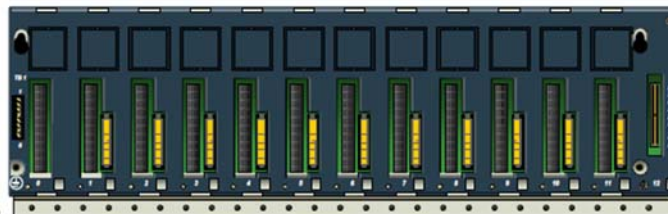


Backplane

Backplane secara umum ada 2 macam :

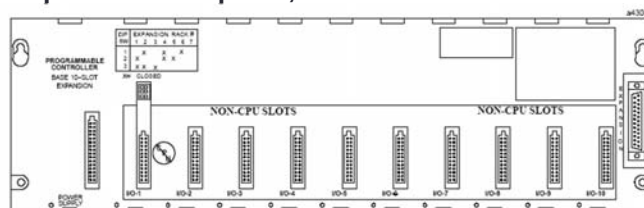
1. CPU Baseplate (Rx3i = Universal) :

- **16 Slot RX3i Universal Backplane, IC695CHS016**
- **12 Slot RX3i Universal Backplane, IC695CHS012**



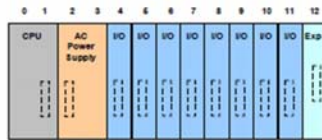
2. Non-CPU Baseplate

- **5 Slot RX3i Serial Expansion Backplane, IC694CHS398**
- **10 Slot RX3i Serial Expansion Backplane, IC694CHS392**





Backplane



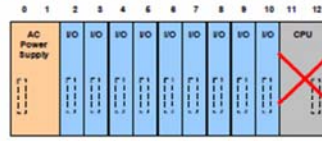
Configured as CPU in slot 0, Power Supply in slot 2



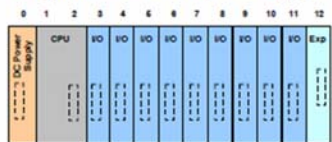
Invalid: AC Power Supply cannot be in Slot 11.



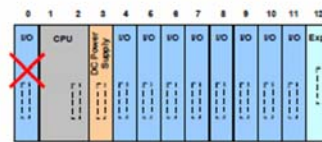
Configured as CPU in slot 0, Power Supply in slot 6



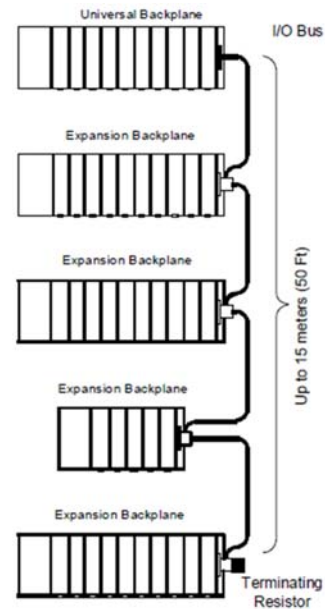
Invalid: CPU cannot be configured in slot 11



Configured as Power Supply in slot 0, CPU in slot 1

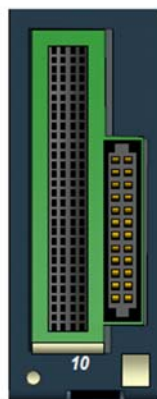


Invalid: Only a Power Supply can be installed in slot 0.

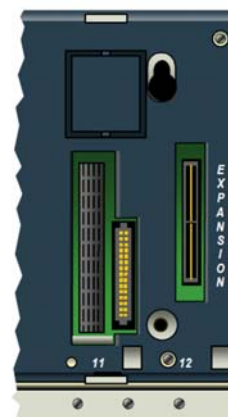


Backplane

Connector
for
PCI Bus



Connector
for
Serial Bus



Slot 1 sampai 11/15

Slot 12/16



Power Supply

Power Supply Type	Catalog Number
120/240 VAC, 125 VDC, 40 Watt Power Supply	IC695PSA040
120/240 VAC, 125 VDC, 40 Watt, Multi-Purpose Power Supply	IC695PSA140
24 VDC, 40 Watt Power Supply	IC695PSD040
24 VDC, 40 Watt Multipurpose Power Supply	IC695PSD140
120/240 VAC, 125 VDC, Serial Expansion Power Supply	IC694PWR321
120/240 VAC, 125 VDC, High Capacity Serial Expansion Power Supply	IC694PWR330
24 VDC, High Capacity Serial Expansion Power Supply	IC694PWR331



IC695PSA040 IC695PSA140 IC695PSD040 IC695PSD140 IC694PWR321 IC694PWR330 IC694PWR331

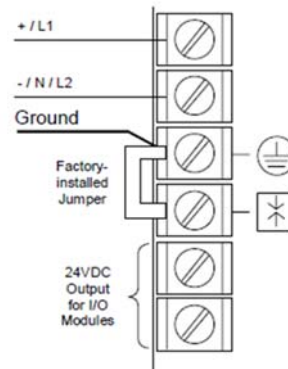
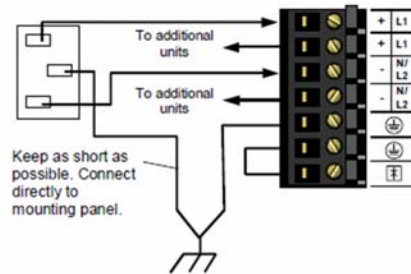


Power Supply



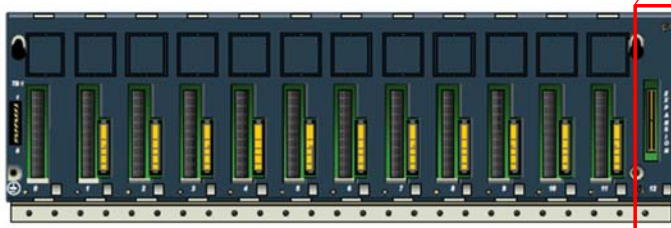


Power Supply



Serial Bus Transmitter

Description	Catalog Number
Serial Bus Transmitter Module	IC695LRE001
Expansion Cable, Wye, 1 meter (3 feet)	IC693CBL300
Expansion Cable, Wye, 2 meters (6 feet)	IC693CBL301
Expansion Cable, 2 Connectors, Built-in Terminating Resistor, 15 meters (50 feet) Length	IC693CBL302
Expansion Cable, Wye, 0.15 meter (0.5 feet) Length	IC693CBL312
Expansion Cable, Wye, 8 meters (25 feet) Length	IC693CBL313
Expansion Cable, Wye, 15 meters (50 feet) Length	IC693CBL314





Digital input Module

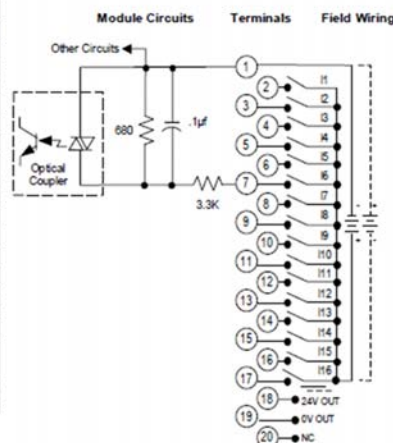
Discrete Input Module	Catalog Number
Input 120 VAC 8 Point Isolated	IC694MDL230
Input 240 VAC 8 Point Isolated	IC694MDL231
Input 120 VAC 16 Point	IC694MDL240
Input 24 VAC/VDC 16 Point Pos/Neg Logic	IC694MDL241
Input 120 VAC 16 Point Isolated	IC694MDL250
Input 120 VAC 32 Point Grouped	IC694MDL260
Input 125 VDC 8 Point Pos/Neg Logic	IC694MDL632
Input 24 VDC 8 Point Pos/Neg Logic	IC694MDL634
Input 24 VDC 16 Point Pos/Neg Logic	IC694MDL645
Input 24 VDC 16 Point Pos/Neg Logic Fast	IC694MDL646
Input 5/12 VDC (TTL) 32 Point Pos/Neg Logic	IC694MDL654
Input 24 VDC 32 point Pos/Neg Logic	IC694MDL655
Input 24 VDC 32 Point Pos/Neg Logic	IC694MDL660
Input Simulator Module	IC694ACC300



Digital input Module IC694MDL645



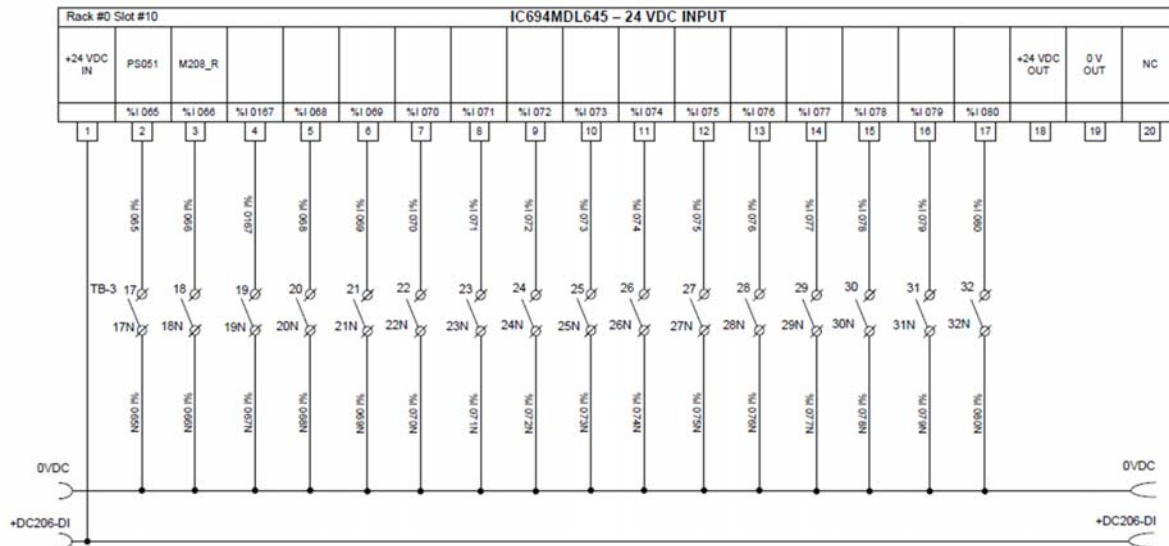
Terminals	Connections
1	Inputs 1-16 Common
2	Input 1
3	Input 2
4	Input 3
5	Input 4
6	Input 5
7	Input 6
8	Input 7
9	Input 8
10	Input 9
11	Input 10
12	Input 11
13	Input 12
14	Input 13
15	Input 14
16	Input 15
17	Input 16
18	24VDC for input devices
19	0V for input devices
20	No connection





Digital input Module

IC694MDL645

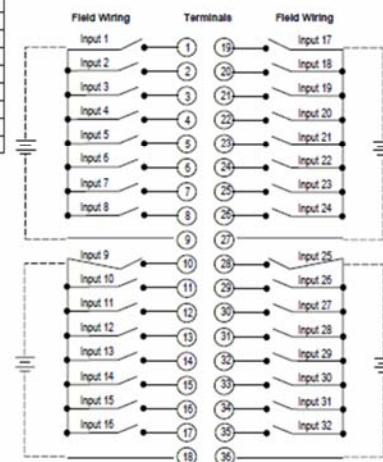


Digital input Module

IC694MDL660

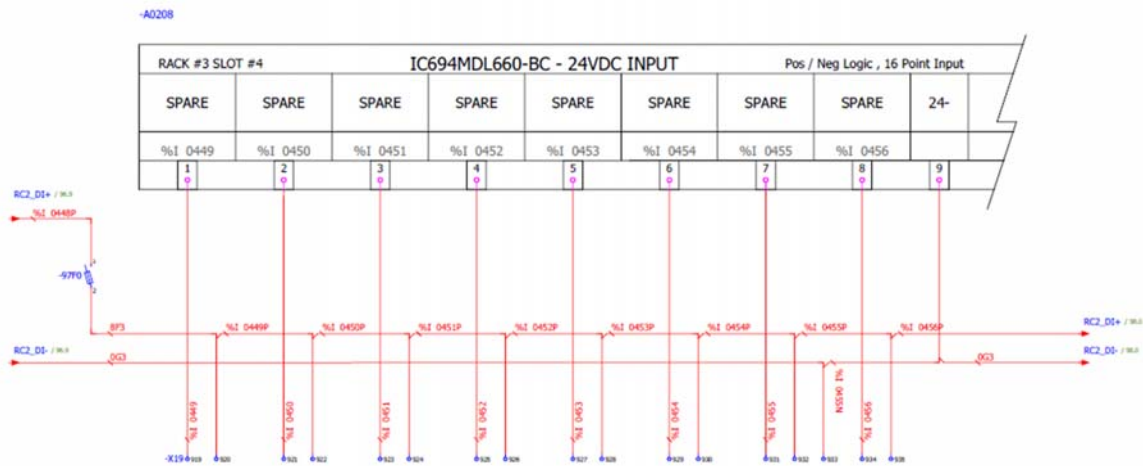


Connections	Terminals	Terminals	Connections
Input 1	1	19	Input 17
Input 2	2	20	Input 18
Input 3	3	21	Input 19
Input 4	4	22	Input 20
Input 5	5	23	Input 21
Input 6	6	24	Input 22
Input 7	7	25	Input 23
Input 8	8	26	Input 24
Common 1 - 8	9	27	Common 17 - 24
Input 9	10	28	Input 25
Input 10	11	29	Input 26
Input 11	12	30	Input 27
Input 12	13	31	Input 28
Input 13	14	32	Input 29
Input 14	15	33	Input 30
Input 15	16	34	Input 31
Input 16	17	35	Input 32
Common 9 - 16	18	36	Common 25 - 32



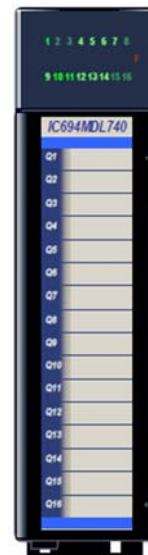
Digital input Module

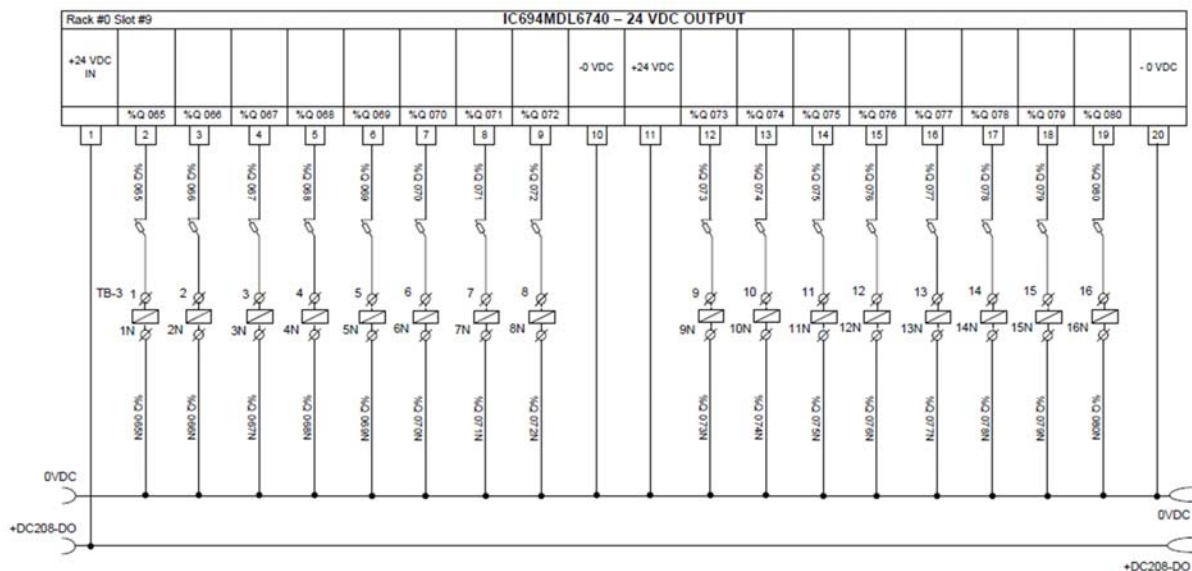
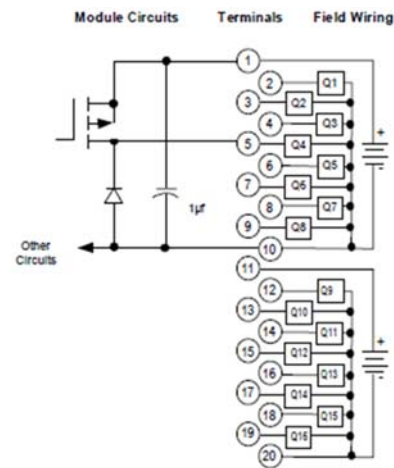
IC694MDL660



Digital Output Module

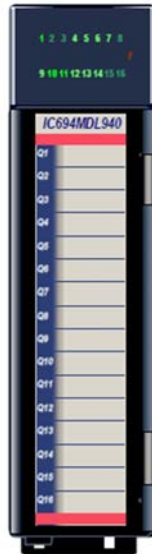
Discrete Output Module	Catalog Number
Output 120 VAC 0.5 A 12 Point	IC694MDL310
Output 120/240 VAC 2 A 8 Point	IC694MDL330
Output 120 VAC 0.5 A 16 Point	IC694MDL340
RX3i Output 124/240 VAC Isolated 16 Point	IC694MDL350
Output 120/240 VAC 2 A 5 Point Isolated	IC694MDL390
Output 12/24 VDC 0.5 A 8 Point Positive Logic	IC694MDL732
Output 125 VDC 1 A 6 Point Isolated Pos/Neg Logic	IC694MDL734
Output 12/24 VDC 0.5 A 16 Point Positive Logic	IC694MDL740
Output 12/24 VDC 0.5 A 16 Point Negative Logic	IC694MDL741
Output 12/24 VDC 1 A 16 Point Positive Logic ESCP	IC694MDL742
Output 5/24 VDC (TTL) 0.5 A 32 Point Negative Logic	IC694MDL752
Output 12/24 VDC 0.5 A 32 Point Positive Logic	IC694MDL753
Output 12/24VDC ESCP 0.75A 32 Point Grouped, Pos.	IC694MDL754
Output Isolated Relay N.O. 4 A 8 Point	IC694MDL930
Output Isolated Relay N.C. and Form C 3 A 8 Point	IC694MDL931
Output Relay N.O. 2 A 16 Point	IC694MDL940



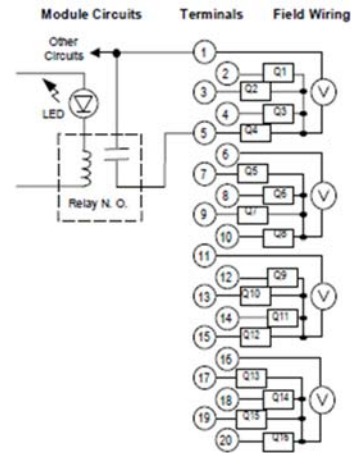


Digital Output Module

IC694MDL940



Terminal	Connection
1	Outputs 1 – 4 common (return)
2	Output 1
3	Output 2
4	Output 3
5	Output 4
6	Outputs 5 - 8 common (return)
7	Output 5
8	Output 6
9	Output 7
10	Output 8
11	Outputs 9 - 12 common (return)
12	Output 9
13	Output 10
14	Output 11
15	Output 12
16	Outputs 13 – 16 common (return)
17	Output 13
18	Output 14
19	Output 15
20	Output 16



Analog Input Module

Discrete Output Module	Catalog Number
Analog Input Module, 4 channel Voltage	IC694ALG220
Analog Input Module, 4 channel Current	IC694ALG221
Analog Input Module, 16/8 channel Voltage	IC694ALG222
Analog Input Module, 16 channel Current	IC694ALG223
Analog Input Module, 8 Channel Non-Isolated / 4 Channel Differential	IC695ALG608
Analog Input Module, 16 Channel Non-Isolated / 8 Channel Differential	IC695ALG616



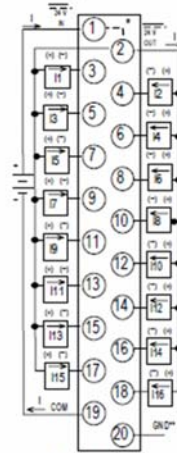
Analog Input Module

IC694ALG223



Terminal	Connection
1	User supplied 24V input; provides loop power via 24VOUT terminal (pin 2)
2	+24V loop power tie point
3	Current Input, Channel 1
4	Current Input, Channel 2
5	Current Input, Channel 3
6	Current Input, Channel 4
7	Current Input, Channel 5
8	Current Input, Channel 6
9	Current Input, Channel 7
10	Current Input, Channel 8
11	Current Input, Channel 9
12	Current Input, Channel 10
13	Current Input, Channel 11
14	Current Input, Channel 12
15	Current Input, Channel 13
16	Current Input, Channel 14
17	Current Input, Channel 15
18	Current Input, Channel 16
19	Common connection to input current sense resistors; user supplied 24V input return or 24VIN return
20	Frame ground connections for cable shields

Field Wiring Terminals Field Wiring



* Internally Connected
 ** Optional Shield Connection

Analog Output Module

Analog Output Module	Catalog Number
Output Analog 2 Channels, Voltage	IC694ALG390
Output Analog 2 Channels, Current	IC694ALG391
Output Analog Current/Voltage 8 Channels	IC694ALG392
Output Analog Current/Voltage 4 Channels	IC695ALG704
Output Analog Current/Voltage 8 Channels	IC695ALG708

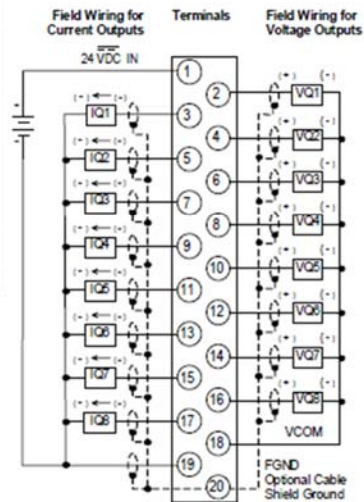




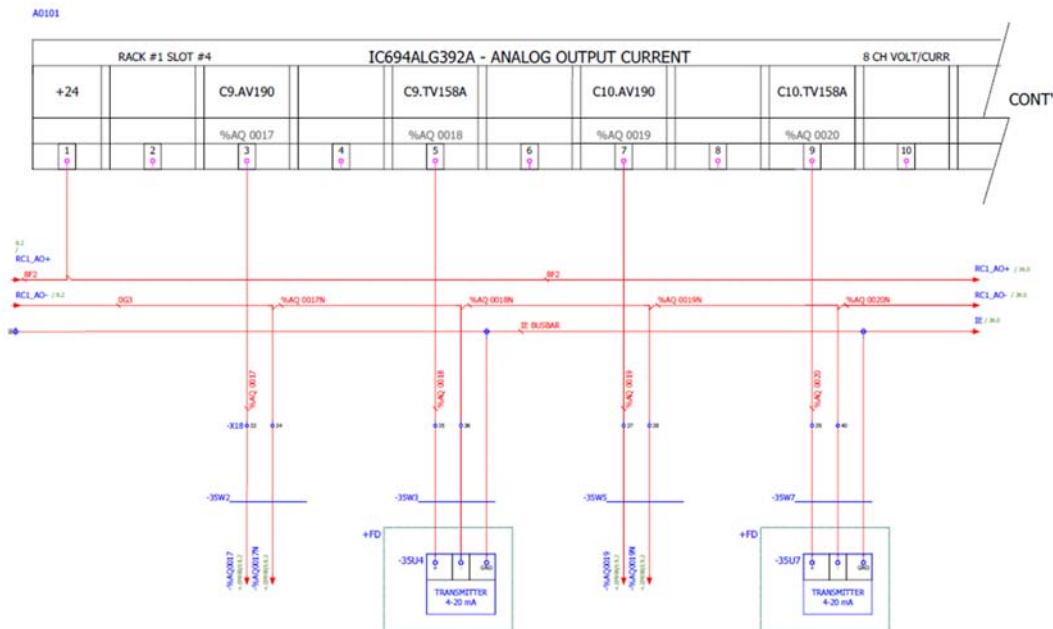
Analog Output Module IC694ALG392



Terminal	Signal Name	Signal Definition
1	24VIN	User Supplied +24 VDC Input
2	V CH 1	Channel 1 Voltage Output
3	I CH 1	Channel 1 Current Output
4	V CH 2	Channel 2 Voltage Output
5	I CH 2	Channel 2 Current output
6	V CH 3	Channel 3 Voltage Output
7	I CH 3	Channel 3 Current output
8	V CH 4	Channel 4 Voltage Output
9	I CH 4	Channel 4 Current output
10	V CH 5	Channel 5 Voltage Output
11	I CH 5	Channel 5 Current output
12	V CH 6	Channel 6 Voltage Output
13	I CH 6	Channel 6 Current output
14	V CH 7	Channel 7 Voltage Output
15	I CH 7	Channel 7 Current output
16	V CH 8	Channel 8 Voltage Output
17	I CH 8	Channel 8 Current output
18	V COM	Voltage Common
19	I COM	Current Common/User +24 VDC Return
20	GND	Frame ground connection for cable shields



Analog Output Module IC694ALG392





Analog Module with HART

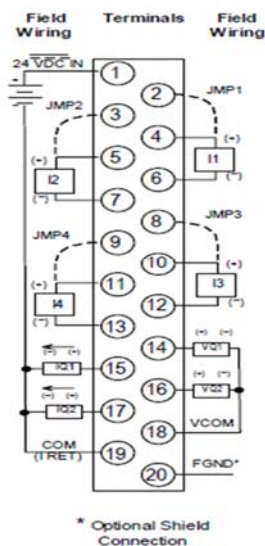
- Analog Input Module, 16/ 8 Channel Voltage / Current, HART: IC695ALG626
- ▪ Analog Input Module, 8/4 Channel Voltage / Current, HART: IC695ALG628
- ▪ Analog Output Module, 8 Channel Voltage/Current with HART: IC695ALG728



Analog Mixed Module

Analog Mixed Module	Catalog Number
Output Module, 4 Inputs, 2 Outputs, Current/Voltage	IC694ALG442

Terminal	Signal	Definition
1	24VIN	User Supplied +24 VDC Input
2	JMP1	Jumper terminal for connecting 250Ω sense resistor for CH1
3	JMP2	Jumper terminal for connecting 250Ω sense resistor for CH2
4	+CH1	Positive connection for differential analog input channel 1
5	+CH2	Positive connection for differential analog input channel 2
6	-CH1	Negative connection for differential analog input channel 1
7	-CH2	Negative connection for differential analog input channel 2
8	JMP3	Jumper terminal for connecting 250Ω sense resistor for CH3
9	JMP4	Jumper terminal for connecting 250Ω sense resistor for CH4
10	+CH3	Positive connection for differential analog input channel 3
11	+CH4	Positive connection for differential analog input channel 4
12	-CH3	Negative connection for differential analog input channel 3
13	-CH4	Negative connection for differential analog input channel 4
14	V _{out} CH1	Voltage output for channel 1
15	I _{out} CH1	Current output for channel 1
16	V _{out} CH2	Voltage output for channel 2
17	I _{out} CH2	Current output for channel 2
18	V COM	Common return for voltage outputs
19	I RET	Common return for User supplied +24 V and current outputs
20	GND	Frame ground connections for cable shields





Special Purpose Module

<i>Module</i>	<i>Catalog Number</i>
Serial I/O Processor Module	IC694APU305
I/O Link Interface Module	IC694BEM320
I/O Link Master Module	IC694BEM321
Genius Bus Controller Module	IC694BEM331
DeviceNet Master Module	IC694DNM200
Motion Mate Module	IC694DSM314
Motion Controller Module	IC694DSM324
Ethernet Interface Module	IC695ETM001
Profibus Master Module	IC695PBM300
Profibus Slave Module	IC695PBS301